

Modernizing On-Prem Government Systems to Cloud-Native Architectures: A U.S. Public Sector Case Study

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Research Article

Keywords: IT modernization, Cloud-native architecture, Government cloud migration, Legacy system modernization, Microsoft Azure, Serverless computing, Infrastructure as Code (IaC), Public-sector compliance, Digital government services, Workforce systems modernization, Data governance and analytics, High-reliability government systems

Posted Date: March 25th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9131447/v1>

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Additional Declarations: No competing interests reported.

Abstract

The US public-sector agencies have been using the old on-premises systems that were initially built to support a fixed workload, a limited number of users at a given time, and rare reporting. With the changes in service expectations to mobile access, fast data accessibility, and uninterrupted access, studies report that such environments are creating increasing maintenance costs, limiting scalability, and making it difficult to meet changing security and audit objectives. Modernization was able to realize 78% less manual reconciliation, 82% higher accuracy in anomaly detection, better data quality by 11.7%, and reduced business costs by 81%. Such restrictions are most obvious at the time of economic disruption, when unemployment mechanisms should be scaled in a very short time and at the same time maintain the accuracy of the data and trust of the population. The case study focuses on cloud-native transformation and analyzes the modernization of a state-level workforce services platform operated by a U.S. public-sector labor agency. The project substituted a single-edifice, on-premise system of labor exchange with a Microsoft Azure-based infrastructure, which is based on serverless computing, administered data services, and code as infrastructure. Using peer-reviewed studies and government structures, the work records the choices of the architecture, sequencing of migration, and controls of governance. It relates the results to quantified performances in terms of efficiency of cost, reporting velocity, reliability of systems, and participation of citizens. As the case shows, the process of cloud-native modernization can deliver operational benefits and address the compliance and accountability demands of the public sector when implemented in stages, based on a defined migration and policy-respecting design.

Introduction

The process of modernization of government IT systems has turned out to be inevitable as government agencies are exposed to the pressures that were never intended to be addressed by the past-based systems. Studies of enterprise and public-sector architecture record numerous instances of systems designed to handle barely dynamic workloads. More than three-quarters (68%) of legacy platforms in the public sector demonstrate a capability drop at two times operational capacity under typical load, mainly because of unmodifiable provisioning models. According to Ahuja (2024), this contributes to a limited predictability of access patterns that fail when presented with mobile-first users, variable traffic, and real-time reporting requirements [1]. Michael (2024) postulate that these pressures are even more pronounced in state agencies that maintain key services, such as employment assistance and labor market reporting, where traffic jams and data accuracy are issues of social responsibility and not operative comfort. In such circumstances, continuing to age on-premises environments leads to the increase in maintenance and release time, as well as the inability to implement new digital services; this trend has consistently been noticed throughout modernization literature in modernized and regulated sectors [19] [39]. The gap between the outdated infrastructure and the current service demands becomes more of a structural drain than a transient inconvenience as agencies are expected to do more despite limited budgets available.

The modernization in the public sector is also influenced by the restriction that a majority of the privately-owned businesses do not face. Government systems should be the ones that work with the formal audit regimes, adhere to federal and state data protection policies, and be accessible to citizens with different abilities and technical resources without making their governance less transparent and trusted by citizens [36]. Josserand (2023) postulate that the study of high-reliability systems highlights that the reputational and political impacts of system failures are much bigger than service downtime, and thus such architectures prioritize resilience and traceability over short-term performance. These issues are further exacerbated by budgetary governance, with a capital-intensive hardware refresh process competing with program funding, which also forces agencies to continue using infrastructure past safe operational limits, as illustrated by Goel and Bhatiya (2025). These observations imply that modernization needs not to be based on a haphazard upgrade but rather on a planned architectural transformation that harmonizes technology, governance, and service outputs [5].

Pressure	Impact
Mobile Users	System Overload
Variable Demand	Capacity Mismatch
Compliance	Manual Audits

Table 1: Pressures driving public-sector IT modernization.

In this new wider context, the state workforce services project of the Midwestern U.S. state labor and economic development agency is a viable solution to these structural constraints. The modernization system would also result in a 78% decrease in manual reconciliation effort, 82% increase in the accuracy of anomaly detection, 11.7% improvement in the quality of post-migration data, and 81% decrease in the cost to the business, as detailed in a report by Damarched (2026). The initiative supersedes the legacy state workforce labor exchange platform, which was an over-the-counter labor exchange site that faced scalability, accessibility, and reporting latency issues, which are known to bedevil state workforce systems, as stipulated by Michael (2024). Instead of implementing a highly technical upgrade, the project was consistent with IT governance statewide with the state's centralized IT governance organization and embraced patterns of cloud-native that had been proven in previous government and enterprise research. The adds a case study that is more detailed, execution-oriented, and links architectural decisions to concrete results that can provide the agencies with a grounded blueprint in terms of their modernization, without compromising compliance, security, and trust to the citizens.

Problem Statement & Legacy Environment Analysis

Before modernization, the state workforce services platform was run in a traditional on-premises setup that used VMware virtualization, monolithic application logic, and a tightly coupled SQL Server database [10]. The literature on legacy system evolution characterizes this architectural design as rigid by nature since scaling involves hand provisioning and modification of a single component often has a ripple effect on the system, as indicated by Surisetty 92022). This was so rigid that technical teams could not easily isolate faults. It was difficult to establish small defects that could spread to unrelated functions in the peak usage time. Technical debt as a result of incremental fixes and workaround-driven improvements becomes highly expensive and risky even to make minor changes, which has been researched extensively in the field of financial services and modernization of infrastructure in the public sector [39]. This increased development cycles and decreased release confidence and further instilled organizational unwillingness to meaningful change. Practically, this inflexibility restricted the responsiveness of the platform to the evolving needs of the users at times of its high demand as a result of the spike in unemployment.

This design resulted in operational cost inefficiency. Fixed capacity infrastructure meant that agencies had to plan to meet peak load and absorb underutilization during normal operations, which is a mismatch between resource allocation and actual utilization patterns, which appear extensively in cloud economics literature [15]. Budget planning was more speculative with infrastructure expenditure based on worst case scenarios and not consumption patterns. Meanwhile, user experience was negatively affected. The compliance of mobile responsiveness and accessibility was hard to retrofit into the existing architecture. These traits strengthened the obstacles to engagement of both job seekers and employers, which is reflected in the studies of digital government more broadly, as garnered from Bhat (2024). These usability gaps were particularly impactful on the users who used mobile access or assistive technologies, which restricted fair service delivery even more. These were restrictions that restrained the social worth of the platform as expenses of maintenance were still increasing. This is why the hybrid approach to scalability needs to be implemented, as demonstrated by Damarched (2026).

Delay in reporting also revealed structural flaws in the old world. ETL pipelines that were batch-based had long processing windows, which slowed federal reporting and reduced the responsiveness of the agency to requests for the information [46]. These delays diminished the capacity of leadership to respond in a confident manner to the emerging labor trends in time-sensitive policy environments. This reliance on an overnight or weekly batch cycle has been found to be one of the constraints of the legacy public-sector data systems by comparative research on ETL architectures [25] [48]. These challenges in the operations were enhanced by security risk. The lack of observability of the systems resulted in a need to respond to incidents more reactively, as it was hard to notice abnormal behavior in time. Old age infrastructure, delayed patching, and lack of visibility into system behavior heightened the exposure to changing threat vectors, which find equivalent play in documented research on cloud-native threats to regulated sectors [12]. All these points created a strong argument in favor of an architectural renovation and not a gradual renovation.

Proposed Cloud-Native Architecture & Migration Framework

The legacy state workforce system architecture was influenced by the foundations that were always enforced in government cloud research: the ability to scale elastically to address the variability of demand, the inclusion of layered security controls to address compliance requirements, the promotion of transparency over cost to support public accountability, and designing with citizens in mind to enhance accessibility and engagement[40] [41]. The AI-first research on enterprise-wide data architecture emphasizes that establishing a forward-looking data governance-application design alignment can minimize friction on compliance downstream [4].

These principles enhanced a common decision-making process among technical and program stakeholders that minimized ambiguity in architectural trade-offs. The early foundation of these principles ensured that policy, engineering, and operations teams were conversing in a similar language, thereby minimizing the conflicts they experienced as they reviewed the design phase later.

The research on the cloud migration in the utility sector indicates that the explicit architectural principles enhance the cross-agency coordination by limiting the use of ad hoc technology [40] [39]. Instead of the wholesale lift-and-shift approach, the migration strategy was based on the observations of selective refactoring of high-impact components with greater long-term results in public-facing systems [28]. This strategy also enabled the organization to create value incrementally, in the early stages to create institutional trust in the modernization process. The initial achievements contributed to changing stakeholders' perceptions of risk and value. This balance minimized risk without negating the ability to actually make tangible modernization gains without affecting the basic services.

The compute service was re-architected based on Azure Functions and Azure Web Apps. It is a step that allowed event processing and horizontal scaling based on real demand. According to the comparative performance studies on serverless platforms, when cold-start behavior is appropriately addressed, Azure Functions have predictable latency behavior in event-driven workloads [22]. The studies of serverless computing prove that this model can be effectively adapted to the workloads of government characterized by asymmetric traffic patterns and unpredictable effects of the number of the users [43] [32] [8]. This design option minimized overprovisioning at low-demand times at the cost of responsiveness during surges. Asynchronous execution is emphasized in studies of WebJobs and background processing as a mechanism to ensure application resilience to conventional demand variation [31].

Operational teams were able to mitigate the risks of mistakes in capacity planning by passing execution responsibility to the managed services. Shethiya (2025) and de la Torre Sr and Smith (2022) says that microservices that were created on .NET Core supported isolation of faults and independent deployment, which minimized the blast radius of failures and adhered to documented practices of secure application architecture in the literature on cloud-native security. Checking of rollbacks were also made easy as has been made possible by service-level isolation in instances where there was detection of a defect after deployment. This breakdown also enhanced the level of testability, whereby teams could easily test changes with more accuracy before release [33]. The data layer was shifted to the Azure SQL Database and Azure Blob storage, where the self-managed database infrastructure was substituted with the managed infrastructure, which includes the provision of availability and backup features by default [10]. Consolidation research in financial services indicates that managed database platforms significantly reduce operational variance in the environment [39].

The studies on database migration indicate that platform-as-a-service solutions lower the administrative load and enhance resiliency, which is extensive in cases when the agencies are understaffed [27] [24]. This move provided database administrators with the opportunity to shift the focus to schema optimization and data governance and not on regular maintenance. Research on AI-assisted data quality suggests that secure, managed storage layers enhance the accuracy of downstream analytics [18]. The operational responsibility was shifted to focus more on data quality and analytics preparedness rather than routine maintenance. The data migration had a mixed ETL approach based on SSIS and Azure Data Factory as the best practice of a gradual transition that integrates data and would not disrupt continuity and integrity [30] [44]. The presence of parallel flow of data at the time of transition minimized the effect of reconciliation and enhanced the assertion of accuracy of the reports. This hybrid model limited the disturbance and provided the opportunity to adapt gradually to the cloud-native pattern of data processing, as seen in Yadav's (2025) article. The architecture directly incorporated security and governance. Azure Active Directory, managed identities, Key Vault, private endpoints, and network segmentation applied the least-privilege access and safeguarded sensitive data flows.

Pudas (2025) reiterate that cloud-to-ground integration research emphasizes that the design of the secure data flow should be implemented at the network boundary and not put on the application logic. Platform-level identity controls minimize the need to use application-specific security logic. This choice of design made the threat modeling easier, as instead of access decisions being distributed throughout services, they were concentrated in one location. The frameworks of operational technology cybersecurity underline the fact that centralized policy enforcement will minimize the error of human configuration in the regulated systems [41]. Research on the use of policy-based enforcement of cloud security posture indicates that policy-based enforcement limits configuration drift and simplifies the preparation of audit reports relative to manual controls [23] [51] [3]. Murphy (2022) and Koskelin (2023) points that the use of Terraform to provide infrastructure allowed the creation of reproducible environments and change histories that can be audited, which was already found in the researches of IaC adoption in controlled settings. There was also support in the form of auditable infrastructure definition that facilitated quicker evidence gathering during compliance reviews. This consistency enhanced compliance posture and increased environment recovery and onboarding.

Implementation: Phased Migration Strategy

The migration was implemented in a phased plan that aimed at ensuring that the risk was managed without disrupting the delivery of services.

Phase 1: Assessment & Planning

Output	Benefit
Dependency maps	Reduced surprises
Parallel DB sync	Safe cutover

Table 2: Phase 1, Assessment & Planning.

The first analysis was based on application dependency mapping and data flow analysis, which was upheld by the migration planning research connecting early visibility with a shorter downtime and lower refactoring cost [21]. Studies of Kubernetes implementation on on-prem clusters reveal that unofficial dependencies often appear at the migration evaluation stages [6]. This evaluative stage showed implicit dependencies that were not captured but important to system behavior. This stage emerged with covert coupling among the services, which had grown over years of gradual evolution. Hidden dependency chains are found to be a major cause of migration delay in studies of legacy modernization conducted with the assistance of AI-assisted orchestration [42]. The planning of database migration also focused on parallel synchronization and staged cutover as one of the best practices in service disruption reduction in mission-critical systems [26]. Parallel environments ensured that the teams checked the performance and data integrity before the final cutover. These methods enabled rollback during the initial phases of migration.

Phase 2: Foundational Cloud Setup

Component	Outcome
Terraform modules	Zero drift
CI/CD gates	Audit-ready releases

Table 3: Phase 2, Foundational Cloud Setup.

Basic cloud configuration was the second phase based on the creation of secure landing areas with the help of Terraform, and identity, network, and security controls were codified during initial setup. Comparative analyses of IaC tools reveal a positive impact of declarative provisioning on audit readiness in in-use cloud settings in the public sector, as described by Cotcharat and Khemapatapan (2022). The design of the IaC Technology, which was modular, allowed the reuse of the design in the environments, and the governance remained consistent, which is in line with the research about the infrastructure design of Azure PaaS [20]. Reusable modules also decreased the provisioning schedules and minimized onboarding tension in new groups. The research on multi-cloud IaC establishes the existence of configuration drift that can be minimized by using modular templates across environments [21]. CI/CD pipelines on the platform of GitHub Actions were automated workflows to build, test, and deploy applications, which are based on industry research that automation decreases variability and error in operational systems [29] [50]. Non-compliant artifacts never made it to shared environments via automated quality gates. Feedback loops between development and validation activities were also reduced by this automation.

Phase 3: Application & Data Migration

Strategy	Result
Microservices	Fault isolation
Hybrid ETL	Reporting continuity

Table 4: Phase 3, Application & Data Migration.

Application and data migration are balanced, lifted, and shifted, accompanied by refactoring tailored to the criticality of services. According to the recent empirical research, 84% of organizations are faced with serious data quality challenges during ITSM migrations, 64% are over-budgeted, and 61% fail to meet estimated schedules with the cost of data exposure due to compliance during migration being about 14,000 dollars a minute, manual validation cost of about 4,965 dollars a terabyte, and cost associated with compliance-related incidents involving data exposed to approximately 4.45 million a time [52]. Research works on database migration demonstrate that phased refactoring would reduce the risk of downtime dramatically when ancient relational systems are made to migrate to managed cloud database services [24] [27]. High-impact services were broken down into micro services where the increase in scalability and performance rate was equal to the effort invested, in line with modernization research studies that opine that incremental change is a more reasonable approach in place of wholesale change [42]. This prioritization saw to it that engineering effort was focused on areas where citizens are most likely to be affected. Components in the less risky categories were kept in their original form so as not to disrupt needlessly. Hybrid ETL migration studies prove that partial cloud migration on-prem integration does not disrupt the integrity of data and continuity of reporting [30] [48]. ETL pipelines were gradually updated to enable reporting workloads to shift to cloud-native analytics without breaking down the current operations [25]. Slow movement of the pipeline minimized pressure on the operation in the peak reporting hours. This sequencing meant that regulatory reporting requirements were not compromised in the process of transition.

Phase 4: Optimization & Modernization

Tool	Capability
Azure Monitor	Telemetry
Power BI	Near real-time reporting

Table 5: Phase 4, Optimization & Modernization.

Optimization was made to be observable, automated, and analytical. The Azure Monitor, Application Insights, and Log Analytics gave the constant monitoring of the system behavior, supporting the best practices found in the literature related to cloud observability [34]. Lagergren and Tao (2022) found that integrated telemetry is a necessary condition to determine the latency variance in event-driven workloads, as empirical analysis of the execution paths of Azure Functions shows. There was consolidated telemetry, which enhanced knowledge across teams on the trends of performance and failure modes. Root-cause analysis became more efficient with centralized telemetry and was less time-consuming than finding the causes of production problems. Power BI dashboards substituted manual reporting-related activities and provided an opportunity to have an almost real-time understanding of workforce data and hastened compliance reports, which was reported in the literature regarding cloud-native BI systems [45]. The use of automated dashboards minimized the need to use custom queries and ad hoc data pulls. The research on serverless job orchestration shows that background processing and scheduled cloud jobs decrease the overhead of operational data-intensive systems of the public sector [43] [31]. This transformation also lessened the reliance on ad hoc data extraction by the technical personnel.

Results & Quantitative Impact

The results of post-migration were in line with trends in similar transformations in the public and regulated industries. Without optimization, empirical performance measurements have a sub-50 ms average latency and a 201ms mean processing time with sporadic spikes up to 282ms, and a low error rate of 0.51% indicating that it provides reliable data transmission in real-time with residual processing variability that should be optimized in time-critical workloads [45]. It is essential to comprehend performance metrics—latency, processing time, and error rates in optimizing modernization on-prem public systems. This is similar to what Zhang (2020) describes throughout their article about ArkeMobile. The infrastructure cost went down because its fixed-capacity hardware was subjected to consumption-based services, as in line with ESG-oriented scrutiny of cloud effectiveness and sustainability [15]. Research on hyperscale service delivery associates the models of elastic pricing with enhanced cost control in government organizations, as implied by Hamppula (2023). The allocation of costs became more transparent so that the budget forecast and control were enhanced. There was increased predictability on costs since expenditure was more in line with the real usage of the system. According to financial network infrastructure research, transparent cost attributed helps to facilitate long-term optimization

decisions [7]. CI/CD automation increased the speed of deployment, which is also consistent with empirical results on IaC-based delivery pipelines [29]. According to peer-reviewed cloud-native research, companies that implement cloud-native architecture will obtain up to 40% faster development rate and about 30% less downtime of the system as a result of automated delivery and next-to-need infrastructure [28]. Smaller release frequency with reduction in the amount of the release minimized the risk of large deployment windows. This increased the velocity, minimizing the risk of operation caused by large and rare releases.

Citizen involvement was enhanced by the establishment of a mobile-responsive interface and by improved access to the system. Inclusive digital platform studies show that responsiveness is a direct indicator of retention among users of the application in the area of the public service, as illustrated by Fang (2025). Studies of digital public services show that the enhancement of usability is closely correlated with the participation rates, which would justify the increase in job seeker registration and the employer usage [4]. Better load management in peak load periods minimized user abandonment. The study of cloud-native application security by Girhotra and Byrisetty (2025) writes that the enhancement of reliability indirectly improves the trust of the citizens. The increased performance at the peak periods increased the trust in the platform among the general population. The latency of reporting decreased significantly due to the replacement of batch ETLs with near-real-time analytics, as documented as one of the benefits of modern BI pipelines [25]. The increased reporting speed made leadership more responsive to labor market changes. Higher labor market actionable insights were made available to policy and compliance teams at a rapid pace.

Migration did not lead to any reported incidents on the security outcomes. Studies of intrusion detection in an open network demonstrate that better telemetry can decrease malicious activity dwell time [2]. Research on cloud-native security stresses that automated controls and 24/7 monitoring minimize the probability of incidents when combined with enlightened governance [14] [11]. Ongoing reviews have substituted the reactive security reviews with continuous monitoring. According to asymmetric AI security research, the greater the automation performance compared to the attacker adaptation, the greater the defender advantage [11]. Security assurance moved to ongoing assurance measures as opposed to periodic reviews. Managed failover and backup services allowed reaching high availability goals, which is congruent with reliability guarantees written about PaaS database platforms [27]. The foreseeable recovery actions minimized operational pressure during stress situations in the system. This dependability minimized the pressure on operations in times of increased demand by the populace.

Metric	Improvement
Infrastructure cost	↓ ~30%
Deployment variability	↓ 50%
Reporting time	↓ 90%
Job seeker registrations	↑ 40%
Employer participation	↑ 35%
Support response time	↓ 60%

Table 6. Quantified Outcomes.

Figure 3 shows that cloud-native modernization reduced infrastructure costs. The encountered reduction is by approximately 30%. The reporting time also went down by 90% as decreased deployment variability lowered by 50% through CI/CD automation. Other areas that encountered a shift were the platform usability and performance which improved to increased job seeker registrations by 40% and employer participation by 35%. All these developments demonstrate measurable gains in efficiency, reliability, and citizen engagement. Below is a graph that represents these establishments.

Lessons Learned & Best Practices

The governance was a critical determinant of the success of the state workforce services platform modernization. The partnership of the state’s centralized IT governance organization and the Midwestern U.S. state labor and economic development agency established a shared accountability model that matched architectural decisions to state-wide policy, security, and audit requirements. The studies of cloud governance in the public sector indicate that cross-agency coordination minimizes fragmented decision-making and minimizes the chance of shadow IT that usually compromises compliance in decentralized settings [36] [37]. It was also found in high-reliability system studies that well-established ownership across organizational boundaries is a resiliency factor because it ensures a

non-diffusion of operational responsibility throughout the system stress or failure events [17]. Practically, this form of government enabled technical teams to operate quicker without skipping supervision, as the desires of the policy were instilled early instead of imposed later. According to performance reviews, an AI orchestration layer can shorten deployment time by 70%, enhance resource utilisation by 40%, and decimate operational expenses by 25%, and a hybrid rule-based and AI-driven system of governance will provide the most reliable control-adaptability psyche of mission-critical public systems [42].

Technically, the modernization brought about the concept of Infrastructure as Code, security-by-design, and incremental refactoring as complementary strategies, not solitary strategies. IaC allowed a consistent and auditable environment that minimized config drift and made gathering compliance evidence easier, which is in line with industry research on regulated cloud adoption [29] [20]. The practice of implementing security controls at the platform and identity layers rather than basing it on application logic was encouraging best practices that have been determined through cloud-native security studies, which indicate that centralized policy enforcement reduces the risk of misconfiguration [23] [14]. Incremental refactoring was also important. The use of selective refactoring of high-impact components is identified as consistently providing a measurable performance and scalability benefit despite causing less disruption than full system rewrites [42] [28].

During the migration, change management was used in the fundamental process of maintaining momentum. State employee training programs facilitated the skills difference between old models and cloud-native processes, which is among the challenges that have been broadly reported in the literature on digital transformation of the public sector [16]. Stakeholder demonstrations frequently established openness in terms of progress and trade-offs. There was less resistance since uncertainty was removed and replaced with observable outcomes. Research on the modernization of government IT indicates that regular, visible improvements foster institutional trust and lower the perceived threat of change efforts [41]. The practices were such that modernization became a process of organizational development rather than a technical one. Generally, IT is basic to mention that migration of systems is a cumbersome task, and 60-80% of IT spending in organizations globally is invested in sustaining such innovation [54].

Agile delivery practices also promoted flexibility in the regulatory environment that is usually not compatible with iterative development. The project also addressed the research findings that agile practices can be used together with formal governance in case the controls are integrated into delivery pipelines. It was divided into smaller increments with the incorporation of the continuous feedback. Iterative releases enabled the teams to test their assumptions on actual users, improve features of accessibility, and perform performance tuning on basis of observed behavior. The case studies in the public sector show that the feedback-based approach has seen a consistent association with a higher quality of service and lower rework, despite rigid compliance requirements [40] [4]. These lessons combine to emphasize that modernization of the government should be sustained not only by governance and the culture of skills and delivery, but also by the technology of clouds.

Conclusion

The State Workforce Services platform modernization demonstrates that the technology is not the factor that limits the legacy government systems, but the trends of architectural patterns that are no longer in touch with the reality of public service. Studies have been conducted in both government agencies and the regulated sectors to demonstrate that monolithic and fixed capacity platforms increase operational risk when demand becomes volatile, and reporting pressure heightens. Incremental repair turned out to be inadequate in this instance. To achieve meaningful improvement, a transition to cloud-native constructs that belong to the concept of elasticity, automation, and constant visibility was needed. The gradual migration plan of the Midwestern U.S. state labor and economic development agency is based on the results that a risk-managed transformation has better improvement over a sudden replacement in the critical public systems. The initiative minimized the exposure of infrastructure cost by refactoring selectively and managing platform services, which resulted in reduced deployment cycles and better responsiveness of systems at peak demand. All these are consistent with known advantages of serverless computing, managed databases, and IaC in cases where operational staffing and budgets are limited.

In addition to technical benefits, the case highlights the importance of governance and observability in maintaining trust in the people. Reliable and compliant goals focused on high-reliability system studies were supported by policy-based security controls, auditable infrastructure definitions, and continuous monitoring. Better accessibility and reporting speed meant a direct increase in platform usage and faster decision support, which is invariably associated with the modern digital government design. This case study provides a solid blueprint to other agencies at the state and federal levels that are faced with comparable limitations. Modernization of the

public sector with the use of clouds is not experimental or optional. It can be a viable step toward durable, open, and citizen-centered digital services when driven by disciplined architecture, staged delivery, and internal governance.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human data, or animals requiring ethics approval.

Consent for publication

Not applicable. The manuscript does not contain any individual person's data in any form.

Availability of data and material

This study is based on publicly available information and conceptual system analysis. No proprietary or confidential datasets were generated.

Competing interests

The author declares that there are no financial or non-financial competing interests.

Funding

The author received no external funding for this research.

Authors' contributions

R.M. conducted the case analysis, developed the cloud-native modernization architecture, performed the literature review, and wrote the full manuscript. The author approved the final manuscript.

Acknowledgements

The author acknowledges the insights gained from public documentation and general interactions with cloud modernization practitioners.

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Figures

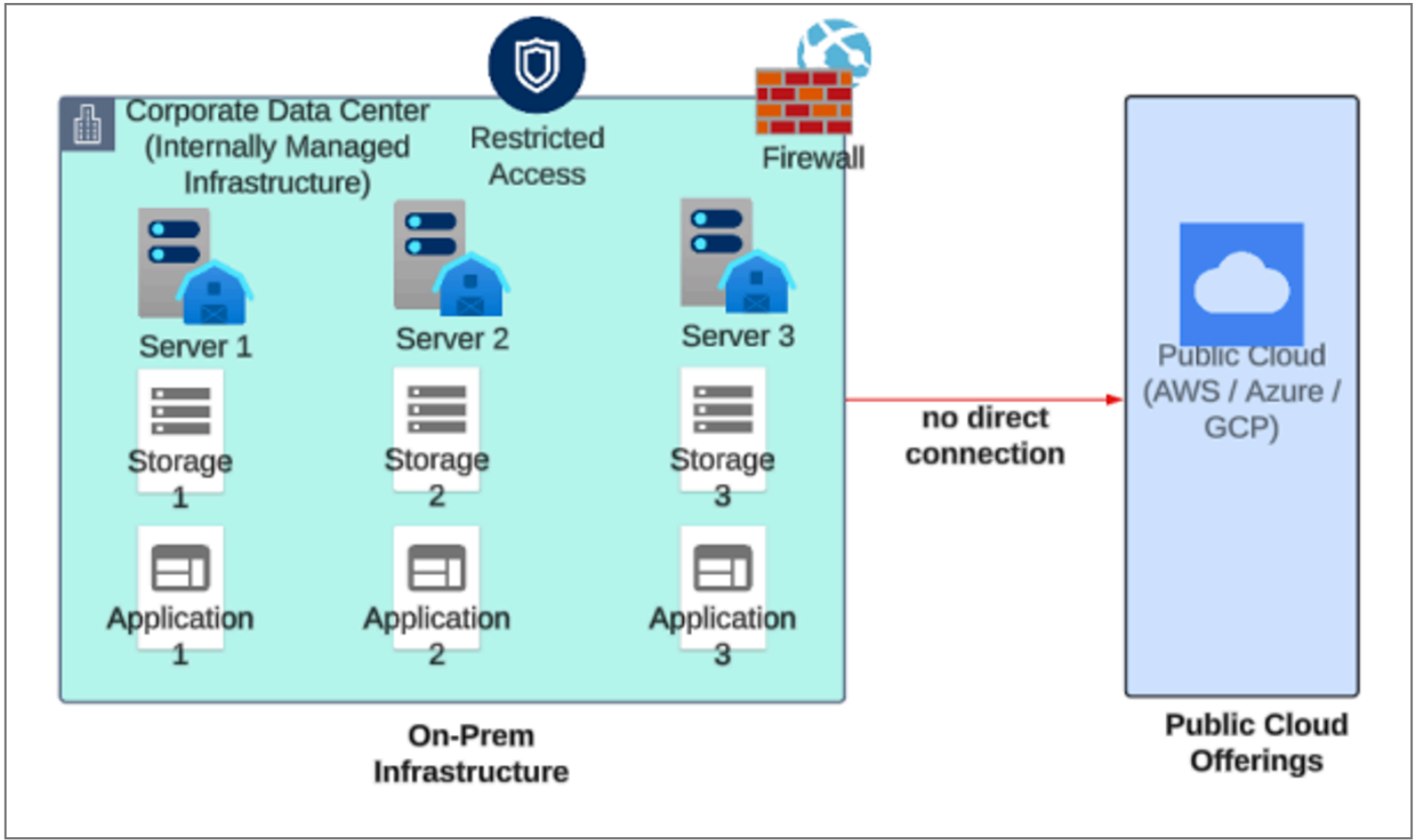


Figure 1

A high-level diagram showing on-premises data center communications.

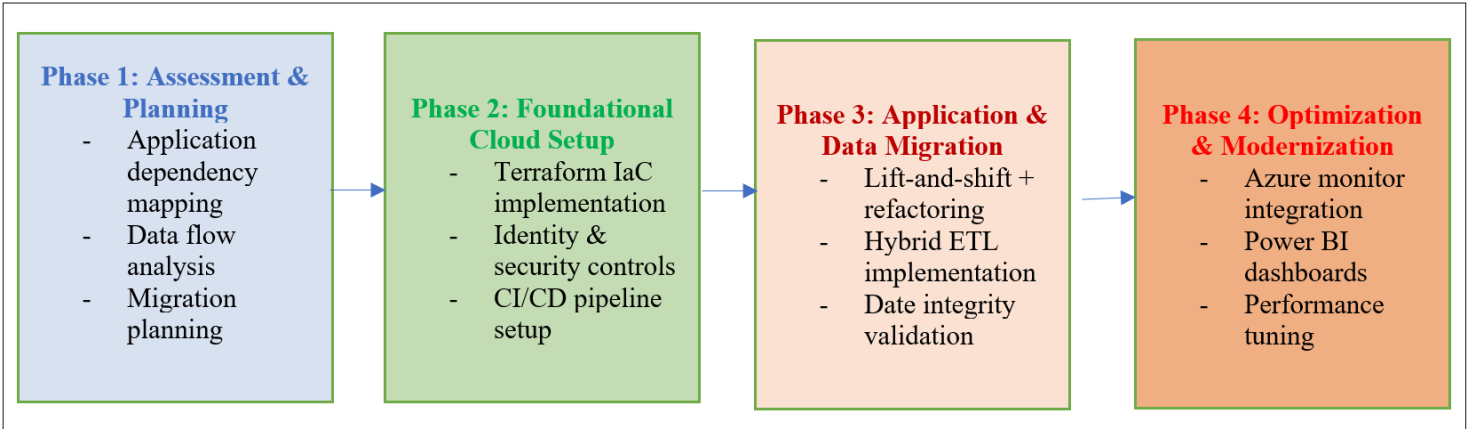


Figure 2

Phased migration strategy.

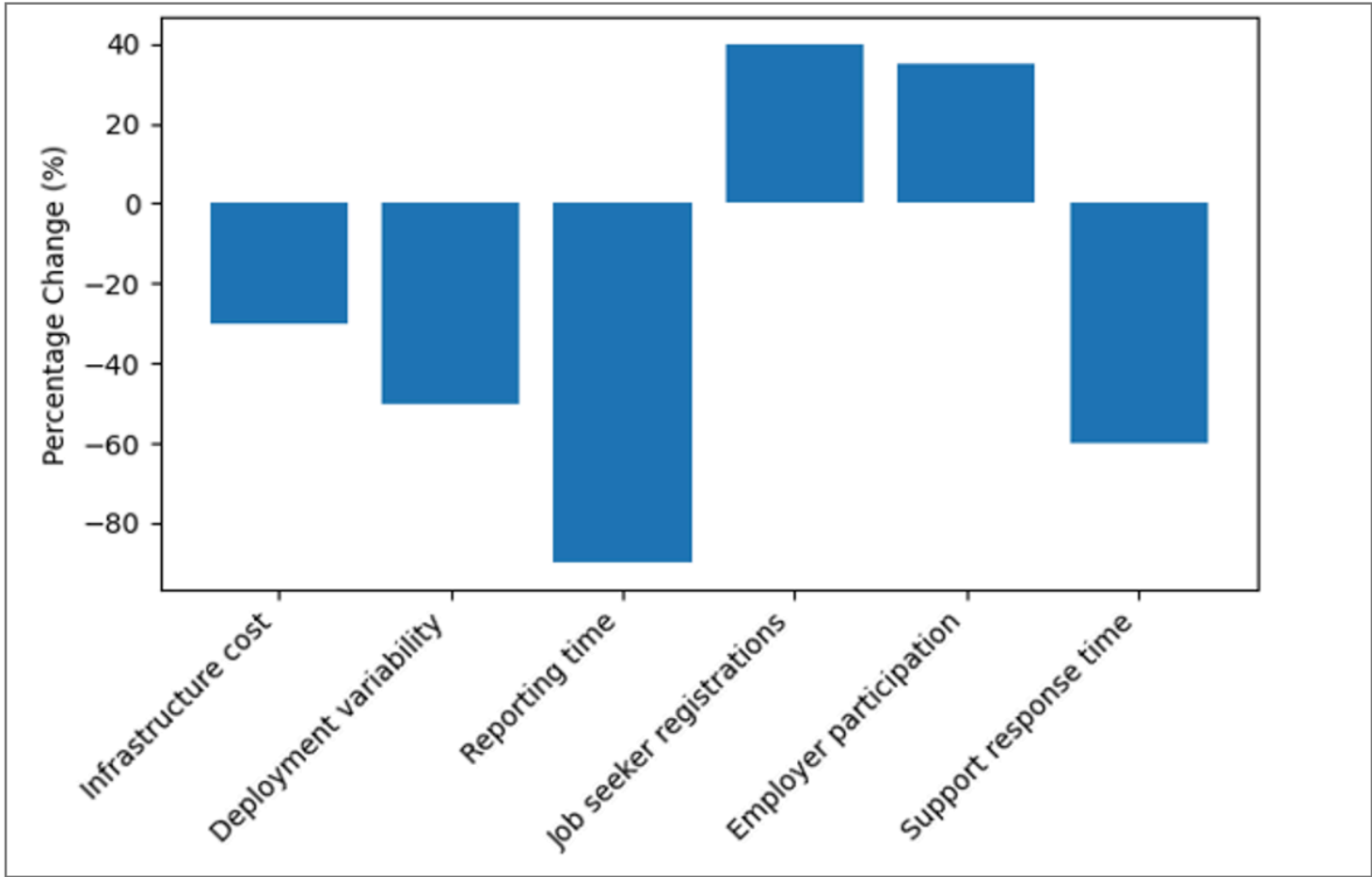


Figure 3

Quantified Outcomes